IntelliCredit: Alternative Credit Scoring and Risk Analytics Report

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COVER PAGE

INTELLICREDIT: ALTERNATIVE CREDIT SCORING AND RISK ANALYTICS

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Declaration

We hereby declare that the project titled "IntelliCredit: Alternative Credit Scoring and Risk Analytics", submitted as part of the requirements for the course 23CSE452 – Business Analytics, is the work carried out by Team 2 under the guidance of our project guide.

We confirm that the project work, analysis and results presented in this report are based on our implementation and that all external sources, datasets and references used have been appropriately acknowledged. We also declare that the report has been prepared in accordance with the institution's academic integrity requirements.

Acknowledgement

We would like to thank our project guide for guiding and supporting us throughout this Business Analytics project. We express our gratitude to the faculty members of Amrita Vishwa Vidyapeetham for providing us with the opportunity and resources to carry out this work.

We also extend our gratitude to our team members for their cooperation, contributions and efforts in data generation, data preprocessing, exploratory data analysis, feature engineering, predictive modelling, time-series forecasting, association rule mining, validation and dashboard development. And also we recognize the developers and providers of the tools, libraries and data sources used in this project, including FRED and the World Bank.

Abstract

Many gig workers have useful digital income records but do not have a normal credit history. This can make it hard for them to get a fair loan. IntelliCredit is a project that studies whether payment behaviour, income changes and digital activity can help estimate default risk.

We created 30,000 synthetic borrower profiles and 360,000 monthly transaction records. We also used real monthly economic indicators from FRED and the World Bank. The project includes data cleaning, charts, machine-learning models, time-series forecasting, association-rule mining and a Streamlit dashboard. The best individual default model reached about 0.74 ROC-AUC on the held-out project data. The time-series model reached a 0.054 percentage-point MAE on the 2024 hold-out period. The association-rule stage kept 93 warning rules after filtering and checked them on unseen borrowers.

We also compared the project with the paper *FinXCredit: An AI-Powered Alternative Credit Risk Assessment System for India's Gig Economy Workers*. Both projects use alternative data for gig workers and use machine learning with explanations. IntelliCredit adds a full validation folder, macroeconomic forecasting, association rules and a clear test against the UCI credit-card dataset. The validation results also show an important limitation: the synthetic data does not transfer perfectly to the UCI data. Therefore, this project should be treated as a research prototype, not as a ready-to-use lending system.

Executive Summary

The main question of this project is simple: **can we use everyday financial behaviour to make a better first estimate of loan risk for people who do not have a strong credit file?**

We built a pipeline with four parts:

1. Clean and understand the borrower and transaction data.
2. Train a model that gives a probability of default.
3. Find combinations of warning signs that often appear together.
4. Forecast how the whole loan portfolio may change during an economic shock.

The project found that a few everyday behaviours carry useful information about loan risk: how steady a person's income is, whether it is falling, whether utility bills are paid on time, how heavy the debt load is, and how long the person has worked on a gig platform. On the project's synthetic test data the model separated safer and riskier borrowers well. But when the model was trained on synthetic data and tested on the public UCI credit-card data, its performance dropped sharply. Good results on synthetic data therefore do not show that the model would work in real lending.

The most useful business idea is to use the model as a second opinion. A lender would look at the score and the warning signs, then make a manual decision. In a simulation of 10,000 applications, a dual-key process (model plus warning rules) approved 71.2% of applicants, cut the default rate among approved loans from 7.87% to 3.40%, reduced default losses by about ₹1.84 crore, and gave a net margin about ₹46 lakh higher than approving everyone. These are simulated results and must be measured in a real pilot.

Our main recommendations are:

5. Use alternative data to support, not replace, human review.
6. Keep the model explainable by showing the main reasons behind a risk score.
7. Test the model on real, approved data before making lending decisions.
8. Recheck the model when economic conditions change.
9. Treat the synthetic-data validation warnings as work that must be improved in the next version.

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Abbreviations

AUC: Area Under the Curve; API: Application Programming Interface; ARM: Association Rule Mining; BNPL: Buy Now, Pay Later; DCR: Distance to Closest Record; DM: Diebold–Mariano test; ECE: Expected Calibration Error; ECL: Expected Credit Loss; FNR: False Negative Rate; FPR: False Positive Rate; FRED: Federal Reserve Economic Data; IQR: Interquartile Range; IV: Information Value; KPI: Key Performance Indicator; KS: Kolmogorov–Smirnov; KYC: Know Your Customer; MAE: Mean Absolute Error; MAPE: Mean Absolute Percentage Error; ML: Machine Learning; PD: Probability of Default; PR-AUC: Precision-Recall Area Under the Curve; RMSE: Root Mean Squared Error; SARIMAX: Seasonal ARIMA with external variables; SHAP: SHapley Additive exPlanations; SMOTE: Synthetic Minority Over-sampling Technique; TSTR: Train on Synthetic, Test on Real; TRTS: Train on Real, Test on Synthetic; UCI: University of California, Irvine; UPI: Unified Payments Interface; VIF: Variance Inflation Factor; WoE: Weight of Evidence.

1. Introduction, Business Objectives and Literature Review

1.1 Context and scope

Traditional credit scoring normally looks at credit-card history, salary slips and previous loans. A gig worker may earn money every month but still not have these documents. This creates a problem for both sides. The worker may be rejected unfairly, while the lender has less information to judge the loan.

IntelliCredit studies alternative signals such as monthly income, income changes, utility payment delays, BNPL amount, savings buffer, UPI activity and gig-work tenure. The project is not meant to make a final legal lending decision. It is a learning project that shows how different analytics methods can work together.

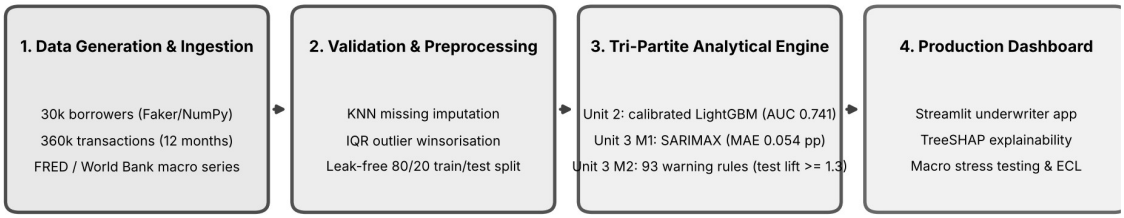


Figure 1. End-to-end IntelliCredit pipeline from data generation to the dashboard. Tool: Python/Matplotlib.

Objective	Measure	Target or check
Understand borrower risk	EDA plots and tests	Clear patterns and no hidden data errors
Predict individual default	ROC-AUC, PR-AUC and calibration	Better than a random model and checked on held-out data
Find groups of warning signs	Support, confidence and lift	Rules must also work on unseen borrowers
Forecast portfolio risk	MAE and MAPE	Compare SARIMAX with simpler baselines
Check whether the pipeline is realistic	TSTR/TRTS, KS, Wasserstein and DCR	Report both good results and mismatches

Table 1. Project objectives and success measures.

1.2 Related paper and research gap

The main paper used for comparison is **FinXCredit: An AI-Powered Alternative Credit Risk Assessment System for India's Gig Economy Workers** by Vikrant Chauhan, Mangesh Sawarkar, Jivesh Nage, Poonam Mishra and Goldi Soni [1]. It was published in IJSREM in May 2026.

The paper describes a large synthetic dataset with more than 100,000 records. It uses an XGBoost model, a five-factor scoring engine and SHAP explanations. It also describes a full web system with KYC, login, OCR and a database. The paper reports an accuracy of 78.6% and ROC-AUC of 0.85 on its own hold-out data.

The paper is useful because it shows that transaction activity and income behaviour can help with gig-worker credit scoring. Gradient-boosted models are also common in credit-scoring benchmarks [2], while SHAP is a standard way to explain model outputs [3]. Our project is similar in its use of alternative data, synthetic data and explainable machine learning. The main difference is that IntelliCredit focuses more on the **full analytics pipeline**: it adds portfolio forecasting, association rules, external macroeconomic data and a separate validation experiment against UCI data.

The comparison must be read carefully. The two projects use different data, labels, feature definitions and test splits. Therefore, their AUC values cannot be treated as a direct competition. A high score on one synthetic dataset does not automatically mean that the model will work on another dataset.

2. Dataset Collection, Validation and Preparation

2.1 Provenance, coverage and challenges

Dataset	Rows	Description	Use
borrower_profiles.csv	30,000	Synthetic borrower records	EDA and default modelling
monthly_transactions.csv	360,000	Twelve months of behaviour per borrower	Features and EDA
macro_india_monthly.csv	108	FRED and World Bank monthly indicators	Forecasting
portfolio_monthly.csv	72	Simulated monthly loan-portfolio history	Forecasting
train.csv	24,000	Stratified training records	Model training and ARM
test.csv	6,000	Held-out records	Final project evaluation
data_compare/UCI_Credit_Card.csv	30,000	Public real-world credit-card benchmark	Pipeline comparison

Table 2. Main datasets used by IntelliCredit and their analytical roles.

The borrower records were generated with Python using fixed seeds. The generator links income, payment behaviour and default risk instead of making every column independent. This makes the data useful for learning, but it also means that some relationships are designed into the simulation.

The primary borrower and transaction data was generated on 9 November 2026 with NumPy, Faker and the project scripts. The project dataset is not copied from Kaggle, GitHub or UCI. The UCI file [5] is used only in the later validation experiment as an external benchmark. The synthetic data has 39 borrower-level fields and 15 transaction-level fields. The target is loan_default_status, a binary indicator for default or 90-day delinquency.

The synthetic generator uses a log-normal income distribution, conditional payment behaviour, bounded loan features, and a logistic default link. The main seeds are 42 for borrower data and machine learning and 2026 for portfolio simulation. API responses are cached so the macro pipeline can also run offline.

Sample row	Age	Income (₹)	Utility reliability	Debt/inflow	Default
BRW-000001	31	26,450	0.92	0.14	0
BRW-000002	39	18,900	0.67	0.31	1
BRW-000003	28	34,120	1.00	0.08	0
BRW-000004	45	22,700	0.75	0.27	0
BRW-000005	26	15,400	0.58	0.39	1

Table 3. Representative final-data sample. The identifiers are synthetic and are included only to show the data shape.

Collection challenges included missing macro observations, different frequencies between annual World Bank [7] indicators and monthly FRED [6] series, and the lack of gig-worker features in the UCI benchmark. The project addresses these with cached fallbacks, documented monthly interpolation in src/macro_api.py, and an explicit feature crosswalk rather than pretending that the fields are identical.

Permitted use: the borrower and transaction records are synthetic and were generated by the team, so no consent from real borrowers was required. FRED [6] and World Bank [7] macroeconomic data are used under their published terms of use, with source attribution. The UCI credit-card file [5] is used under the repository's terms for public datasets and is cited as an external benchmark only.

2.2 Data quality, privacy and ethics

The cleaning pipeline checks missing values, invalid categories, outliers, date alignment and target leakage. Structural nulls, such as a missing default month for a loan that never defaulted, are kept because they have a valid meaning. The project does not use real borrower names or private account data.

The project also records a correction for a date bug. Earlier code used a 30-day time step, which could create duplicate month labels. It was changed to `pd.date_range(..., freq="MS")`, which gives correct calendar months.

Check	Finding	Action and final status

Check	Finding	Action and final status
Missing values	Portfolio default months are structurally null for non-defaulted loans	Retained with documented meaning
Duplicate months	Earlier 30-day increment caused month collisions	Replaced with calendar-month dates
Invalid categories	Possible unmapped ordinal values	Centralised maps and assertions
Class imbalance	Default rate is about 7.87%	Use PR-AUC, recall and calibration, not accuracy alone
Leakage	SMOTE before cross-validation was identified as unsafe	SMOTE moved inside the training pipeline
Privacy	No real borrower PII is stored	Synthetic names and IDs only

Table 4. Data quality, privacy and ethics checks.

The project has no consent requirement for real borrowers because the primary records are synthetic. If real data is added later, the team would need documented authorization, purpose limitation, access control, retention rules and compliance review under India's Digital Personal Data Protection Act, 2023. The main representation bias is that the synthetic population is designed around gig workers and may not represent salaried borrowers, older borrowers or every Indian region. The UCI mapping also uses imputed constants for fields that do not exist in that dataset.

2.3 Transformations and usable dataset

The raw files are checked and transformed before analysis. Numeric missing values use median or KNN-style imputation where appropriate; categorical values use mode or explicit ordinal maps; extreme values are winsorised using IQR limits; dates are converted to monthly timestamps; and transaction rows are aggregated to borrower-level features. Model scaling, encoding and SMOTE are model-specific and are fitted only on training folds.

Transformation	Before	After	Reason
Borrower records	30,000 raw	30,000 usable	No rows removed for normal synthetic values

Transformation	Before	After	Reason
Monthly transactions	360,000 raw	360,000 usable	Aggregated by borrower and month
Train/test split	30,000	24,000 / 6,000	Stratified 80/20 split
ARM data	24,000 borrowers	19 binary behaviour flags	Makes itemsets comparable
Portfolio data	250,191 loan-month rows	72 monthly portfolio rows	Aggregate for forecasting

The final field mapping is:

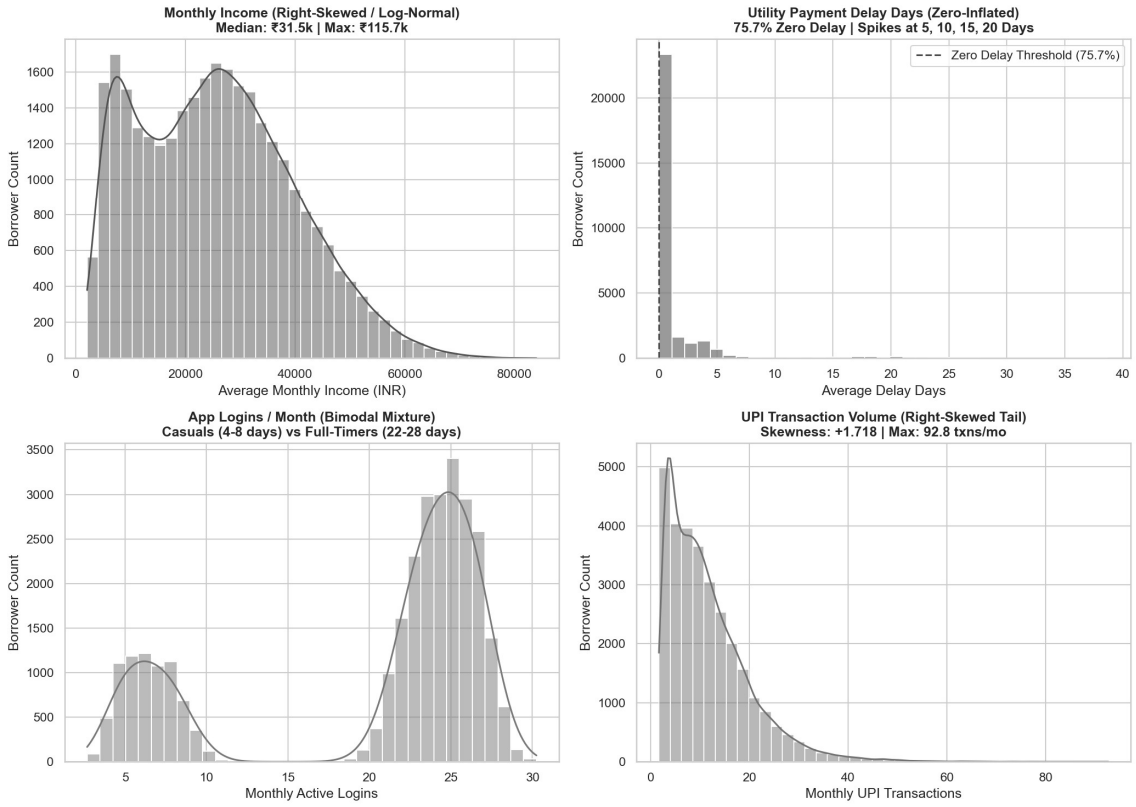


Figure 2. Shape validation distributions. Tool: Python/Matplotlib; source: `validation/step4_fidelity_audit.py`.

The validation plots help us see whether the generated data looks reasonable and whether it is too close to the comparison data.

The plots show that some features are reasonably close, especially age and the income proxy. Other features are clearly different. For example, utility reliability is much higher in the synthetic data, while the UCI data has more payment delay behaviour. This does not make the project data useless, but it means that transfer results must be reported honestly.

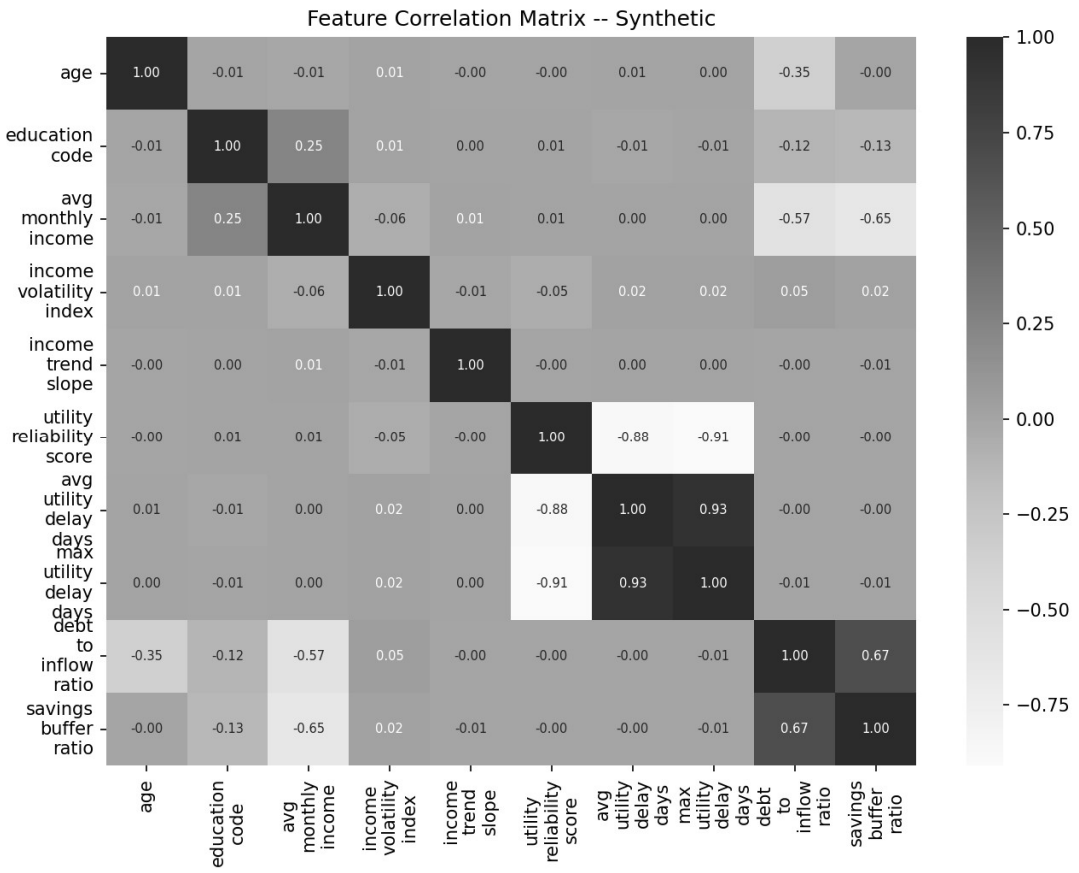


Figure 3. Correlation structure in the synthetic data. Tool: Python/Seaborn; source: validation/step4_fidelity_audit.py.

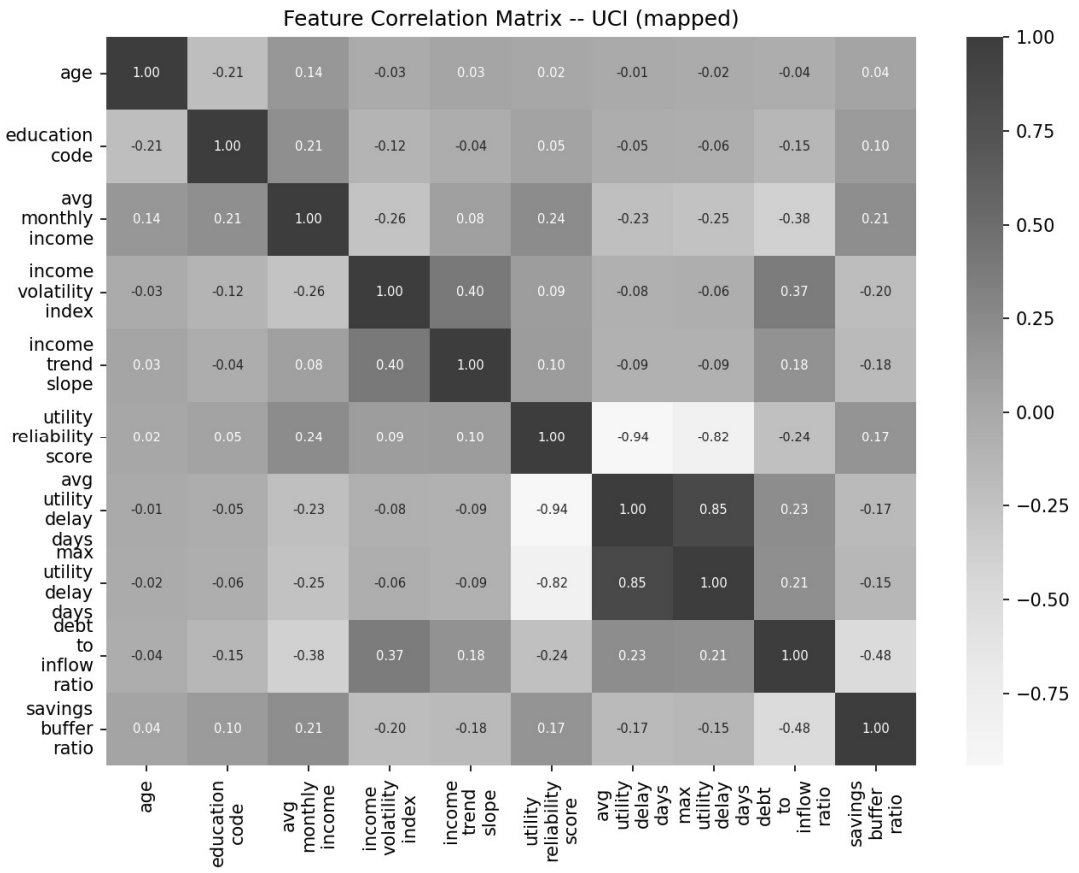


Figure 4. Correlation structure in the mapped UCI data. Tool: Python/Seaborn; source: validation/step4_fidelity_audit.py.

The two correlation plots also look different. The correlation-matrix Frobenius distance is 2.5595. This supports the conclusion that the two datasets are not the same type of population.

Field or group	Preparation	Used in
Income and transaction values	Cleaning, winsorisation and aggregation	EDA, ML, ARM
Utility payment fields	Reliability and delay features	EDA, ML, ARM
Gig tenure	Numeric and grouped versions	EDA, ML, ARM
Loan default status	Kept as target only	ML and ARM validation
Portfolio monthly data	Monthly default-rate series	SARIMAX
Macro indicators	API collection and cached fallback	SARIMAX
UCI credit-card fields	Feature crosswalk to project schema	Pipeline validation

Table 5. Final field mapping across Units 1–3.

3. Exploratory Data Analysis and Visualisation

EDA was used before modelling so that we could understand the data instead of treating the model as a black box. The notebooks check distributions, groups, correlations and default rates.

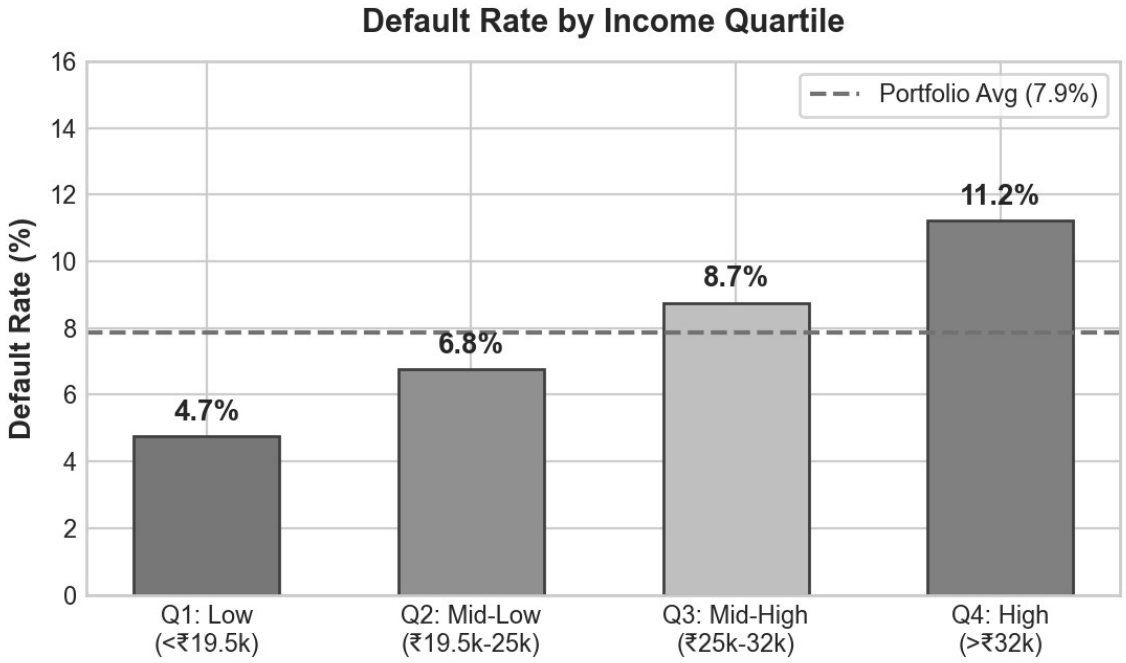


Figure 5. Exploratory borrower patterns from the project data. Tool: Python/Matplotlib; source: notebooks/02_M2_exploratory_data_analysis.ipynb.

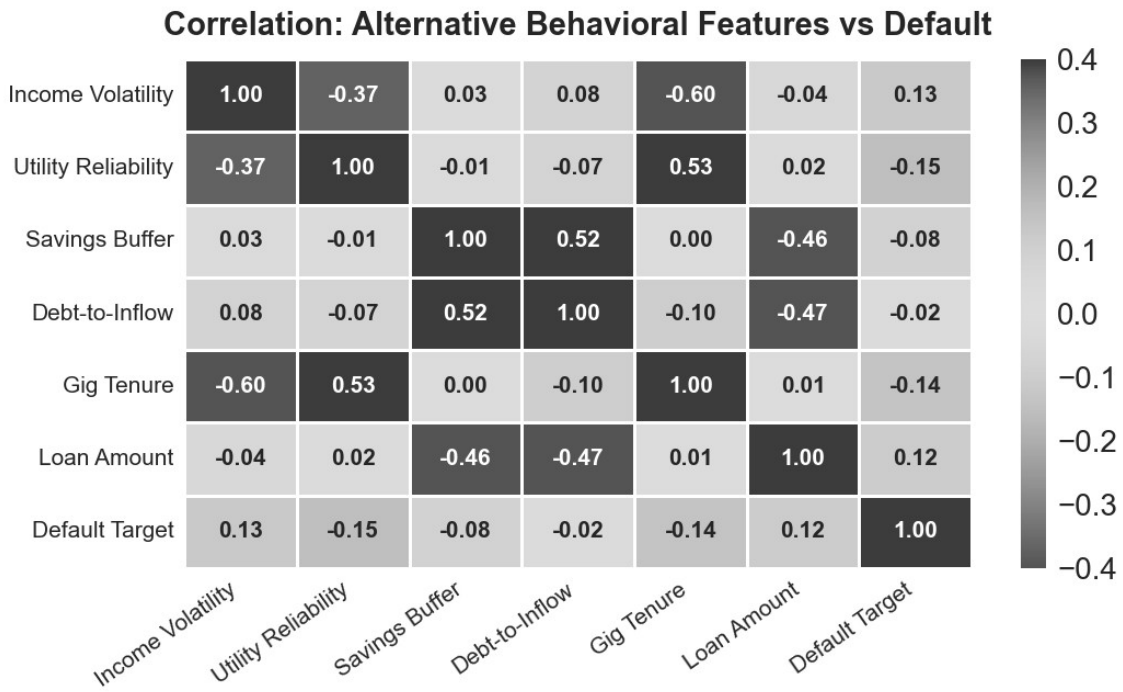


Figure 6. Additional behaviour and default comparisons. Tool: Python/Seaborn; source: notebooks/02_M2_exploratory_data_analysis.ipynb.

The main patterns were:

- Borrowers with short gig-work tenure generally show higher default risk.
- Higher income volatility is linked with more defaults.
- Utility payment delays are a useful warning signal.
- A larger debt-to-income ratio is linked with higher risk.
- The default class is smaller than the non-default class, so accuracy alone is not a good measure.

The project uses ROC-AUC and PR-AUC together with recall and calibration. This is important because a model can get high accuracy by predicting "no default" for almost everyone.

3.1 Important findings and hypotheses

The EDA supports four hypotheses. First, shorter gig tenure is associated with more defaults because the income history is less stable. Second, high income volatility is associated with lower repayment ability. Third, utility delays and BNPL burden can work as early warning signals. Fourth, macroeconomic stress can increase portfolio defaults even when individual borrower features stay the same. Mann–Whitney and chi-square tests in the EDA notebook are used as evidence for group differences, but they are interpreted as associations and not causal effects.

4. Feature Engineering and Predictive Modelling

4.1 Predictors, target and feature construction

The project creates features such as income volatility, income trend, utility reliability, savings buffer, debt-to-inflow ratio, loan-to-income ratio and repayment ratio. These features are easier to understand than a long list of raw monthly columns.

The target is `loan_default_status`. Predictors are borrower demographics, aggregated transactions, loan request information and derived financial ratios. VIF, mutual information, RFE and WoE/IV checks are used to remove unhelpful or highly repeated signals. PCA and t-SNE are used for understanding structure, not as the only basis for a lending decision.

4.2 Model development and leakage audit

The data is split into 80% training and 20% testing. SMOTE and other preprocessing steps are placed inside the training pipeline so that the test set is not used to create synthetic training examples. This is one of the most important leakage checks in the project.

Five-fold cross-validation is used inside the training data for tuning. The held-out 6,000 borrowers are used once for final comparison. The leakage audit checks that the target is excluded, entity rows do not cross the split, scalers and imputers are fitted within folds, and SMOTE is not run before the folds are created.

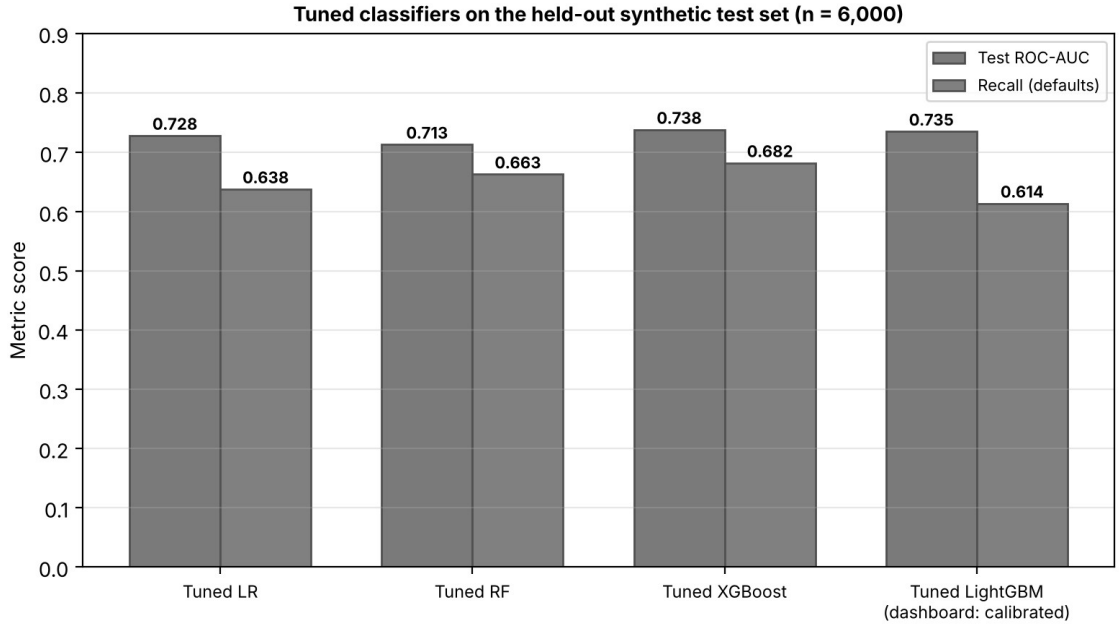


Figure 7. Tuned classifiers on the held-out synthetic test set (values from Table 6). Tool: Python/Matplotlib.

Model	Test ROC-AUC	Test PR-AUC	Recall
Logistic Regression	0.728	0.197	0.638
Random Forest	0.713	0.178	0.663
XGBoost, tuned	0.738	0.204	0.682
LightGBM, tuned	0.735	0.201	0.614
Calibrated LightGBM	0.741	0.199	Checked with calibration

Table 6. Project model results on the held-out synthetic test set.

The simple logistic model is the baseline. The tuned tree models are the proposed alternatives, and calibrated LightGBM is the dashboard choice because its output is intended to be read as a probability.

The calibrated LightGBM model was selected for the dashboard because it gives probabilities that can be used in a break-even decision rule. XGBoost is also useful for ranking applicants. SHAP explanations show the main reasons for a prediction, such as gig tenure, income trend, utility reliability and income volatility.

These results are good for the project data, but they are not proof that the model is ready for a bank. The external validation in Section 8 gives a more cautious picture.

4.3 Interpretation, error analysis and subgroup limitations

SHAP is used to explain which features push a prediction higher or lower. The strongest project signals are gig tenure, income trend, utility reliability, loan-to-income ratio and income volatility. These explanations are useful for a reviewer, but SHAP shows association with the model output; it does not prove that changing a feature would change a borrower's real risk.

Subgroup checks show where the model is least reliable. At the cost-optimal cutoff ($P^* = 0.114$), the false-negative rate, meaning creditworthy applicants who are rejected, is 45.6% for metro borrowers and 60.6% for rural borrowers, and 40.5% for postgraduates compared with 61.1% for borrowers with below-10th-pass education. Removing location and education inputs left test AUC unchanged at 0.7406 and lowered the rural false-negative rate to 57.6%, so the recommended scoring policy drops those inputs. A future real-data study should still report subgroup AUC, recall and approval rates using actual outcomes.

5. Advanced Analytics

5.1 Method 1: Time-series forecasting

The first advanced method asks: **when might portfolio defaults rise?**

The project uses 72 months of simulated portfolio data and macroeconomic indicators such as inflation, unemployment, interest rates, GDP growth and exchange rate. SARIMAX was compared with seasonal naive, SARIMA, Holt-Winters and gradient boosting.

The SARIMAX model gave a 2024 hold-out MAE of 0.054 percentage points and MAPE of about 16%. The macro variables help the model react to conditions that are not visible in the past default-rate series alone. The result is a forecast, not a guarantee.

Model	2024 MAE / RMSE (pp)	2024 MAPE (%)	Diebold–Mariano vs SARIMAX
SARIMAX with macro drivers	0.054 / 0.068	16.36	—
Gradient boosting (lags + macro)	0.065 / 0.073	21.91	p = 0.068
Holt-Winters exponential smoothing	0.075 / 0.102	23.86	p = 0.003
Univariate SARIMA (1,1,1)(1,0,1)12	0.088 / 0.102	29.54	p = 0.007
Seasonal naive benchmark	0.124 / 0.164	40.44	p < 0.001

Table 7. Time-series forecast errors on 2024 hold-out actuals (12 months) and Diebold–Mariano tests against SARIMAX.

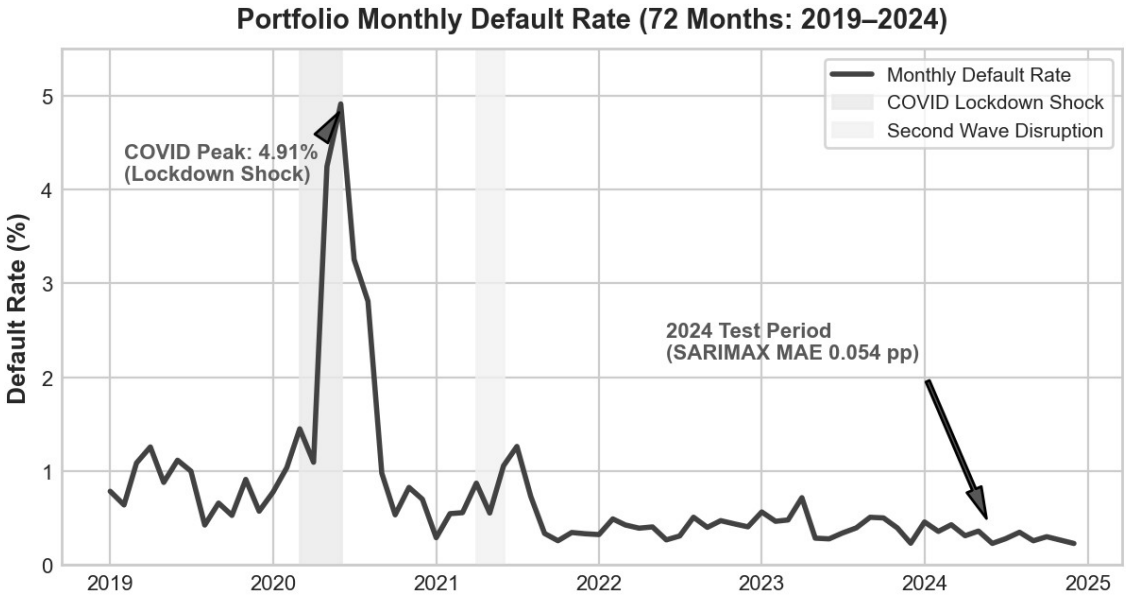


Figure 8. Historical portfolio default-rate trend and modelled time-series behaviour. Tool: Python/Matplotlib; source: `notebooks/06_R2_M1_time_series_forecasting.ipynb`.

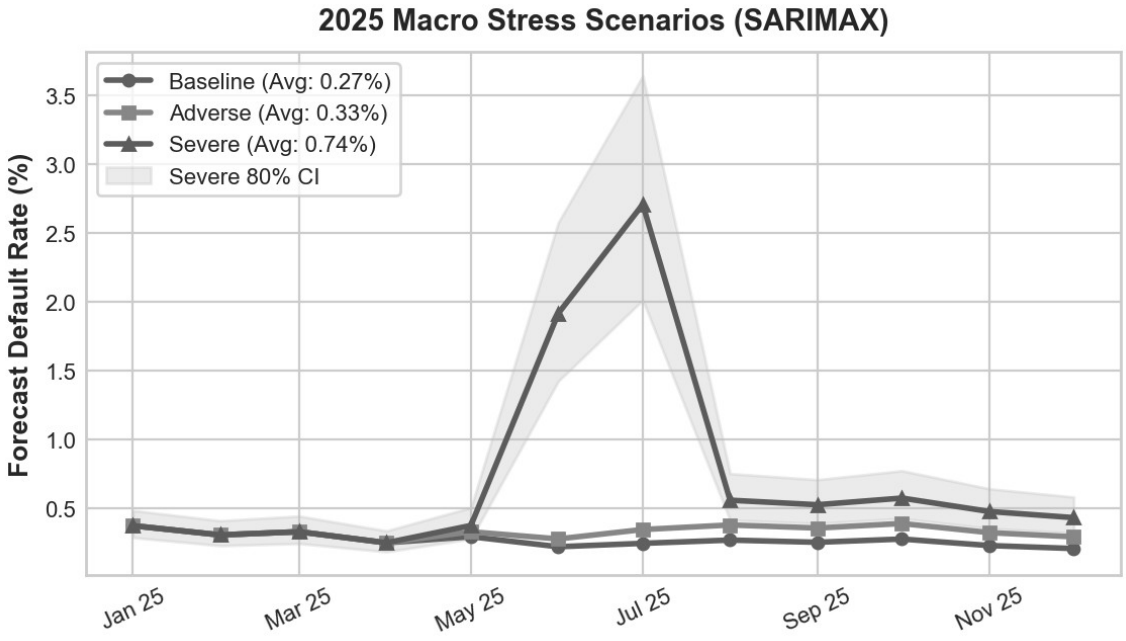


Figure 9. Baseline, adverse and severe portfolio scenarios. Tool: Python/Matplotlib; source: `models/ts_forecast_scenarios.csv`.

5.2 Method 2: Association rule mining

The second method asks: **which warning signs appear together?**

Nineteen behaviour flags were converted into itemsets. Apriori and FP-Growth produced frequent itemsets. Rules were then filtered using support, lift, statistical testing and redundancy checks. Rules were kept only if training confidence was at least 15%, training lift at least 1.5, and a Bonferroni-

corrected Fisher exact test gave $p < 0.05$. The final project output contains 93 warning rules. They were checked on the held-out borrowers, and the project reports that all 93 kept test lift above 1.3.

This method is useful because it gives simple statements such as "declining income + volatile income + low savings buffer is a risky combination." It does not replace the ML score; it gives an extra explanation for a manual reviewer.

Warning rules triggered	Default rate (%)	Relative to no rules	Risk group
0	4.9	1.0×	Baseline
1–2	7.2	1.5×	Low risk
3–5	14.8	3.0×	Moderate
6 or more	24.7	5.0×	Severe risk

Table 8. Default rate by number of warning rules triggered on held-out borrowers (values shown in Figure 10).

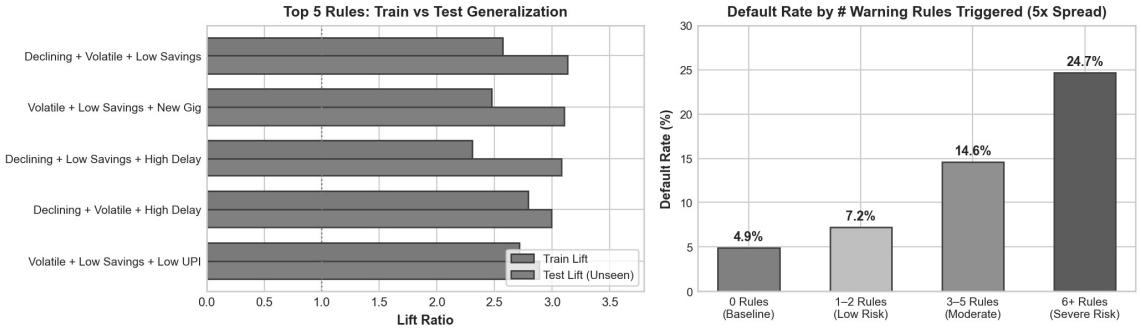


Figure 10. Association-rule support, confidence and lift. Tool: Python/Matplotlib; source: `notebooks/07_R2_M2_association_rule_mining.ipynb`.

The main limitations are that frequent patterns are not automatically causal, and rules can change when the population or threshold definitions change. The test-set lift check reduces this risk, but it does not remove it completely.

6. Comparative Analysis and Business Impact

6.1 Comparing the two advanced methods

Time-series forecasting works at the portfolio level. It helps answer how the whole book may change during an economic shock. Association rules work at the borrower level. They help explain combinations of warning signs.

The methods do not use the same metric and should not be ranked against each other. Their value is that they answer different questions and can be shown together in the dashboard.

Each method has a clear strength and limitation. Forecasting gives portfolio-level stress estimates from macro data, but it cannot say which borrower will default. Rule mining gives statements a reviewer can read, but its rules depend on thresholds and can change if the borrower population changes.

6.2 Comparison with FinXCredit

Topic	FinXCredit paper	IntelliCredit
Main problem	Credit scoring for Indian gig workers	Credit scoring plus portfolio risk
Main data	More than 100,000 synthetic records	30,000 borrower profiles and 360,000 transactions
Main model	XGBoost and five-factor heuristic	Logistic Regression, RF, XGBoost, LightGBM and calibrated model
Explainability	SHAP	SHAP plus human-readable warning rules
Macro forecasting	Not a main part of the paper	SARIMAX with FRED and World Bank data
Application	FastAPI, React, KYC and login features	Streamlit research dashboard
External pipeline test	Reported hold-out test	UCI crosswalk, TSTR/TRTS, KS, Wasserstein and DCR
Main caution	Synthetic-data results	Strong synthetic-to-UCI mismatch is reported

Table 9. Comparison with the reference paper. Numerical AUC values are not directly comparable because the datasets and testing setups differ.

The FinXCredit paper is stronger in product features such as authentication and KYC. IntelliCredit is stronger in combining several analytics methods and showing a separate pipeline validation. Neither project should claim production readiness without testing on real approved-loan records.

6.3 Business impact and scenarios

The dashboard can use a break-even probability rule. In the project scenario, a normal portfolio has an estimated expected credit loss of about ₹1.59 crore. An adverse scenario increases this by about ₹36 lakh, while a severe scenario increases it by about ₹2.74 crore. These are modelled scenarios, not measured future losses.

The practical use is to help a lender decide when to:

- send more applications to manual review;
- reduce loan size for unstable income;
- keep extra loss reserves during a stress period; and
- explain why an application needs more information.

Scenario	Assumption	Estimated result	Type
Best / baseline	Normal macro conditions	₹1.59 crore ECL	Modelled scenario
Adverse	Unemployment +1.5 pp and GDP decline	₹1.95 crore ECL	Modelled scenario
Severe	COVID-style stress multiplier	₹4.32 crore ECL	Stress scenario

Table 10. KPI-based business scenarios. These are modelled estimates, not measured future losses.

Table 11 summarises a simulation of 10,000 applications under four underwriting policies. Net margin equals gross interest income minus default loss. These are simulated results, not measured outcomes.

Underwriting policy	Approval rate / default rate (%)	Default loss (₹ crore)	Net margin (₹ crore)
Naive baseline (approve all)	100.0 / 7.87	2.66	+2.14
Strict traditional bureau proxy	42.0 / 3.38	0.48	+1.53
ML score alone (cost-optimal)	74.7 / 4.40	1.11	+2.47
IntelliCredit dual-key engine	71.2 / 3.40	0.82	+2.60

Table 11. Simulated outcomes per 10,000 applications under four underwriting policies.

These estimates depend on the simulated portfolio size, loss assumptions and macro overlay. The operational cost is mainly manual-review time, data access and model monitoring. The expected benefit is lower exposure to poorly understood risk, but it should be measured in a real pilot rather than assumed.

Recommendations are ranked by simulated impact and feasibility. First, use the dual-key review, which gives the largest net margin and the largest fall in default losses at moderate operating cost. Second, hold extra loss reserves during macro stress, which protects against the severe scenario at low implementation effort. Third, reduce loan size for unstable income, which lowers default risk but also reduces approved volume and therefore ranks last.

7. Interactive Application, Figure Index and Reproducibility

7.1 Mandatory dashboard or website

The Streamlit dashboard has tabs for borrower underwriting, portfolio trends, warning rules, dataset exploration and combined insights. It reads the saved model, portfolio outputs and rule tables. Some outputs are precomputed, while the applicant score and filters are calculated when the user interacts with the dashboard.

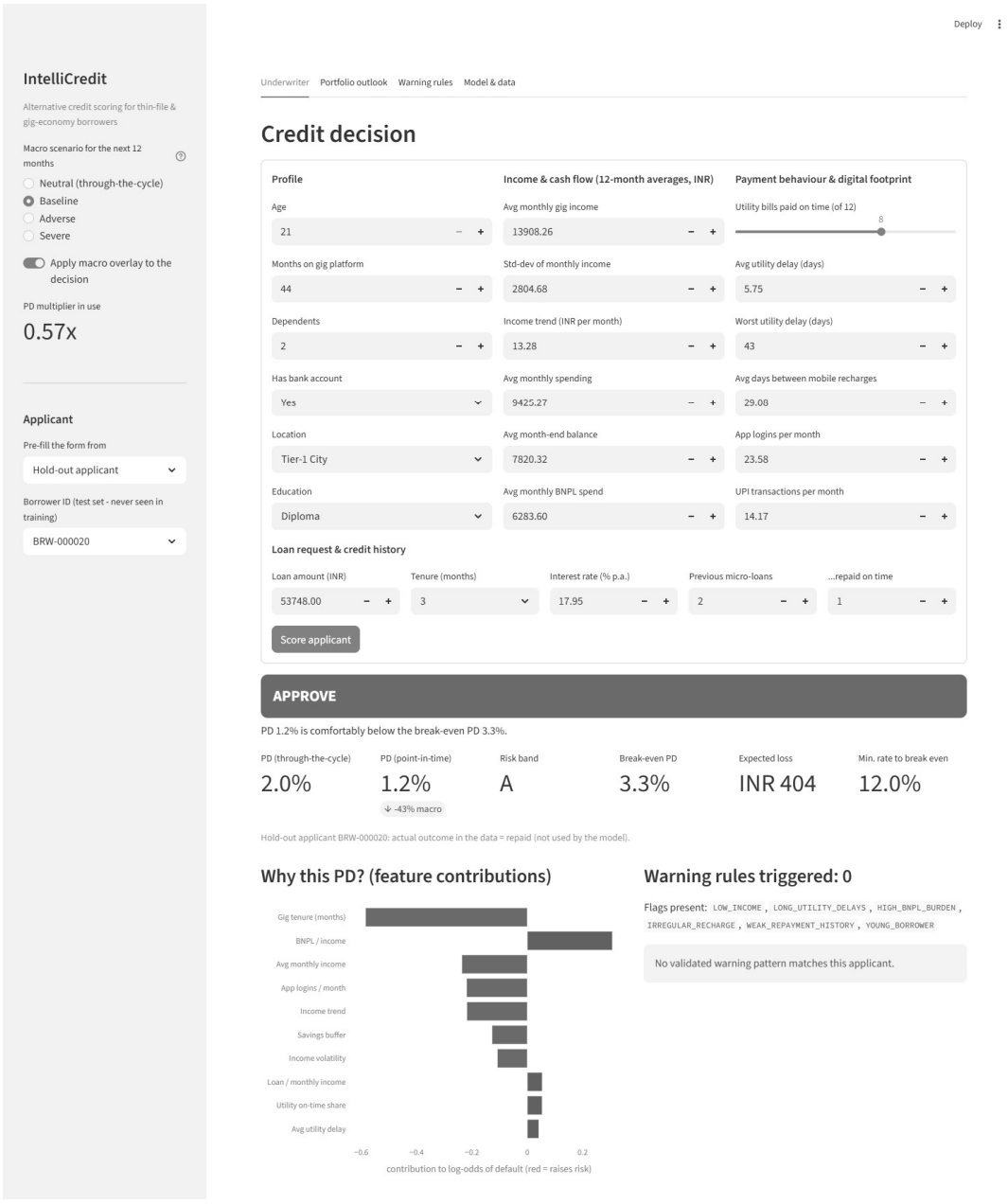


Figure 11. Underwriter dashboard view with a borrower risk result. Tool: Streamlit (dashboard/app.py); screenshot.

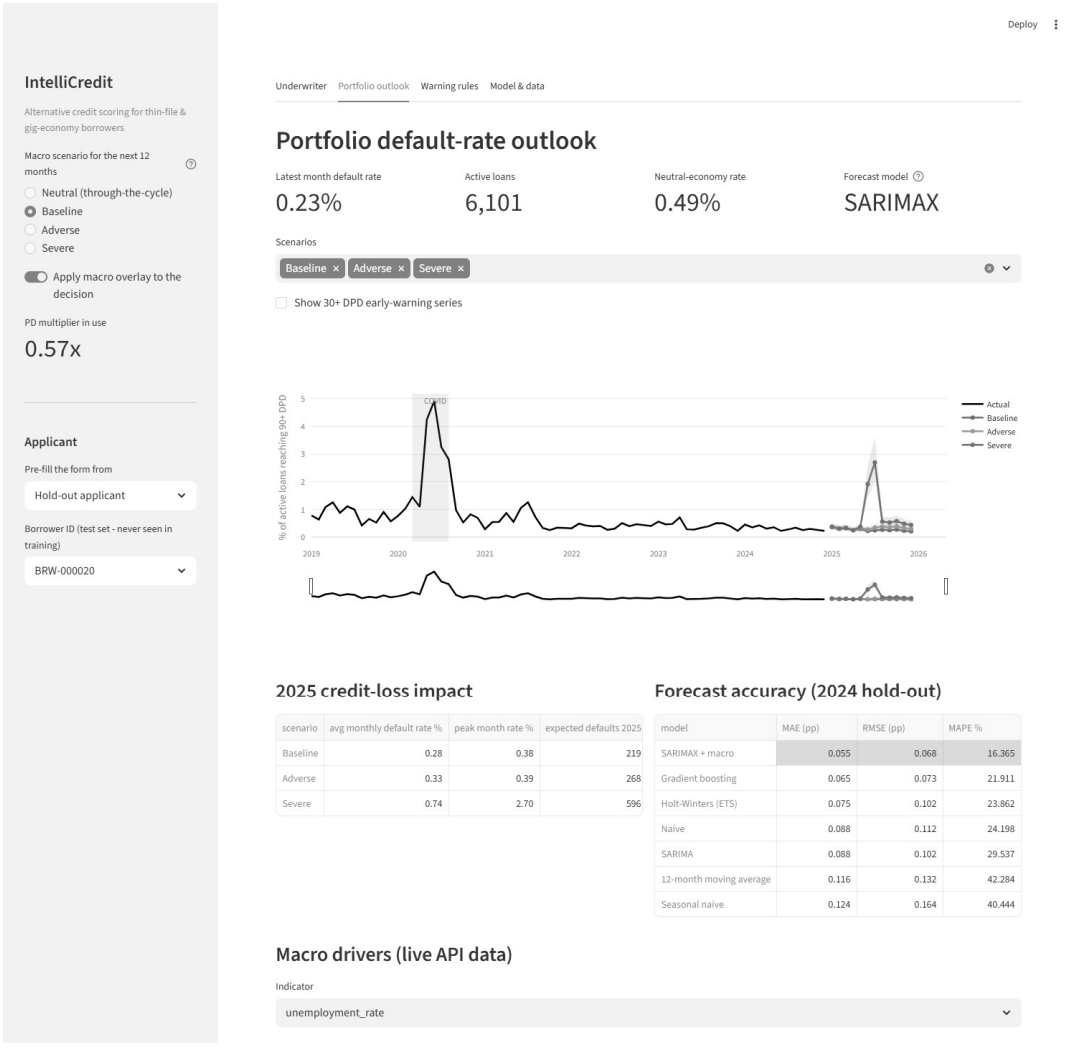


Figure 12. Portfolio view for trend and scenario analysis. Tool: Streamlit (dashboard/app.py); screenshot.

To reproduce the project:

```
python -m venv venv
pip install -r requirements.txt
python data_generation.py
python -m src.macro_api --offline
python -m src.portfolio_simulation
python -m src.underwriting_model
python validation/run_all.py
streamlit run dashboard/app.py
```

The repository includes notebooks, source code, data dictionaries, model files, reports and validation outputs. The fixed random seeds make the main experiments repeatable.

The dashboard was checked by loading the application, opening each tab, scoring a sample borrower, changing portfolio filters and viewing the warning-rule table. Known limitations are that it depends

on the saved model files, the macro API may be unavailable offline, and it is a research prototype rather than a secure production service.

7.2 Figure index and reproduction map

Figures are embedded once in the analytical sections. This index gives the source and purpose without repeating the images.

Figure	Title / tool	Section and purpose	Source
1	Pipeline architecture / Python	Workflow connecting Units 1–3 and dashboard	`reports/presentation_figures/pipeline_architecture.png`
2	Shape distributions / Python	Synthetic versus UCI feature shapes	`validation/step4_fidelity_audit.py`
3–4	Correlation matrices / Seaborn	Relationship structure comparison	`validation/results/fidelity_corr_*.png`
5–6	EDA charts / Matplotlib	Behaviour and default patterns	`notebooks/02_M2_exploratory_data_analysis.ipynb`
7	Model comparison / Matplotlib	Held-out model metrics	`notebooks/04_M4_predictive_modeling.ipynb`
8–9	Forecast and scenarios / Matplotlib	Portfolio trend and ECL stress	`notebooks/06_R2_M1_time_series_forecasting.ipynb`
10	Association rules / Matplotlib	Support, confidence and lift	`notebooks/07_R2_M2_association_rule_mining.ipynb`
11–15	Dashboard and validation plots	Application and transfer checks	`dashboard/app.py`, `validation/results/`

Table 12. Figure index and reproduction map.

7.3 Code and environment

The repository is `https://github.com/Darun1017/intellicredit-report`. It contains `data/`, `data_compare/`, `data/processed/`, `notebooks/`, `src/`, `models/`, `validation/`, `dashboard/`, `reports/` and `requirements.txt`. Python 3.10+ is recommended. The requirements file lists the analytics, machine-learning, visualization and Streamlit packages.

The project uses only synthetic borrower data for its main analysis. The UCI file is retained as a validation reference, with its source and mapping documented in `validation/step1_feature_crosswalk.py`. No secrets are stored in the repository. Macro API data is cached and may be regenerated with the offline fallback.

8. End-to-End Validation, Traceability and Contributions

8.1 Completion and validity

The declared scope is complete: Unit 1 covers cleaning and EDA; Unit 2 covers feature engineering and predictive modelling; Unit 3 covers SARIMAX and association rules; and the Streamlit dashboard connects the outputs. The main threats to validity are synthetic-data circularity, mismatch between synthetic and UCI populations, imputed fields in the crosswalk, and the lack of real lending outcomes.

Objective / KPI	Analysis or method	Result / metric	Evidence	Recommendation
Understand borrower risk	EDA and plots	Behavioural differences identified	Figures 5–6; Notebook 02	Use behaviour as supporting evidence
Predict default	ML with held-out test	Calibrated LightGBM AUC 0.741	Figure 7; Notebook 05	Recalibrate on approved real data
Find warning combinations	Apriori / FP-Growth	93 rules; test lift ≥ 1.3	Figure 10; Notebook 07	Use as manual-review signals
Forecast portfolio risk	SARIMAX	MAE 0.054 pp	Figures 8–9; Notebook 06	Monitor macro stress
Validate transfer	TSTR/TRTS and fidelity	TSTR AUC 0.5941; gap 0.2827	Figures 13–15; `validation/`	Do not deploy without real-data testing

Table 13. Objective-to-evidence traceability matrix.

8.2 Project tracking and individual work

The project was tracked through GitHub Issues and Projects using five stages: **Backlog, To Do, In Progress, Review/Testing and Completed**. The repository issue history and the project-management file record the task sequence from data generation to dashboard and final report. The board reference is documented in `GITHUB_BOARD_SETUP.md`; the task evidence is in `reports/project_management_and_contributions.md`.

Team member	Assigned work	Evidence

Team member	Assigned work	Evidence
Darun R G (CB.SC.U4CSE23215)	Data generation, feature engineering, ARM preparation and validation	`data_generation.py`, notebooks 01/03, `data/DATASET_VALIDATION_SUMMARY.md`
VIMALHARIHAR S K (CB.SC.U4CSE23254)	Model tuning, SARIMAX forecasting, association rules and synthesis	notebooks 04/06/07/08, models/
JUTURU NAGA ABHINAVA SAI (CB.SC.U4CSE23563)	EDA, statistical tests and portfolio simulation	notebook 02, src/portfolio_simulation.py
Gowtham Reddy Peddireddy (CB.SC.U4CSE23322)	Cleaning, Streamlit dashboard and HTML report	notebook 01, dashboard/app.py, reports/interactive_dashboard.html
D R Adithyaa Seyyone (CB.SC.U4CSE23305)	Model evaluation, SHAP, recommendations and reporting	notebook 05, reports/

Individual marks can differ according to participation and ability to explain the work. Commit and issue history provide additional evidence.

Table 14. Individual contribution records.

8.3 AI use and originality declarations

Generative AI was used for code suggestions, debugging, plain-language editing and report formatting. The team decided the data-generating process, features, experiments, validation tests and conclusions. Team members ran the code and corrected errors. In particular, the transfer failure was kept in the report instead of being hidden.

The report uses cited ideas, algorithms, software and external sources. No copied personal data or restricted borrower records are included.

8.4 Pipeline validation using `data_compare` and `validation`

The `data_compare` folder contains the UCI Credit Card Default data used as a real-world comparison. The `validation` folder runs four steps:

10. **Feature crosswalk:** maps UCI columns into the IntelliCredit feature format.
11. **TSTR/TRTS experiments:** trains on one data source and tests on the other.
12. **Calibration and policy check:** checks probability reliability and break-even decisions.
13. **Fidelity audit:** compares distributions, correlations and record distances.

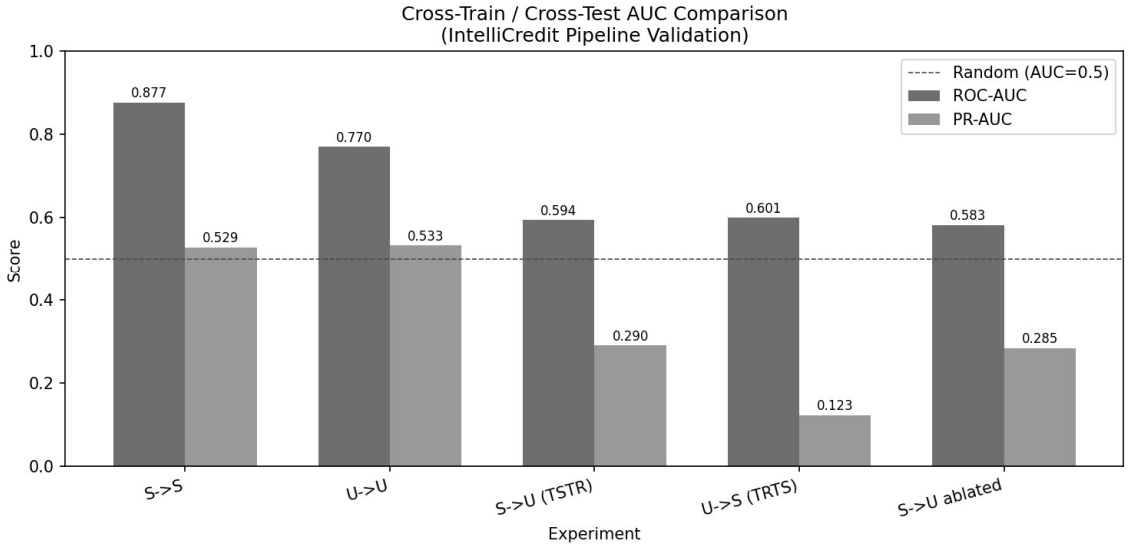


Figure 13. AUC results for same-domain and cross-domain experiments. Tool: Python/Matplotlib; source: validation/step2_tstr_trts.py.

Experiment	AUC	PR-AUC	Meaning
Synthetic → Synthetic	0.8768	0.5290	Strong same-domain result
UCI → UCI	0.7700	0.5330	Real-data benchmark result
Synthetic → UCI (TSTR)	0.5941	0.2904	Weak transfer to mapped real data
UCI → Synthetic (TRTS)	0.6005	0.1234	Weak reverse transfer

Table 15. Pipeline validation results from `validation/results/run_summary.txt`.

The synthetic-to-UCI AUC gap is 0.2827, labelled “strong domain mismatch” by the validation script. This is the most important limitation in the report. It tells us that the project model learns patterns that do not fully match the mapped UCI population.

8.5 Calibration, policy and fidelity checks

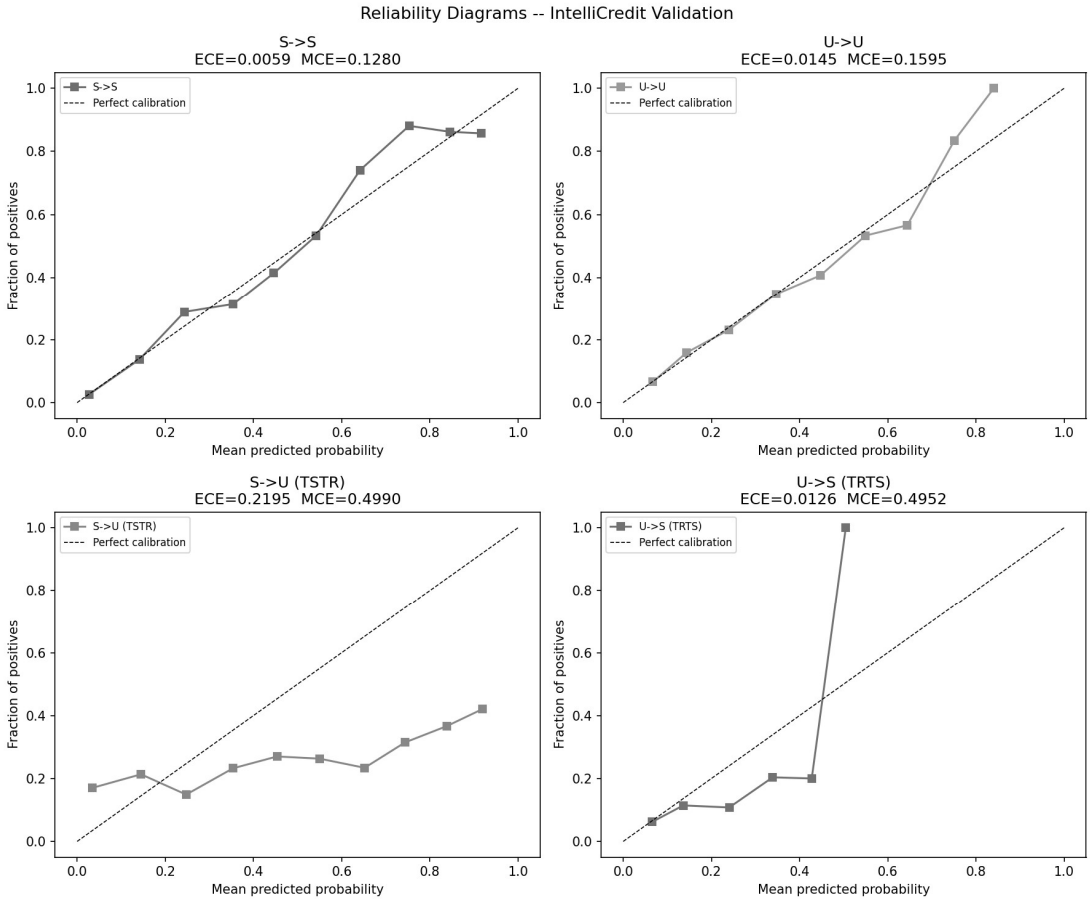


Figure 14. Reliability diagrams for the transfer experiments. Tool: Python/Matplotlib; source: validation/step3_calibration_policy.py.

The same-domain synthetic result has a low ECE of 0.00594, but the TSTR result has a much higher ECE of 0.21952. This means the probability values are not reliable when moved directly to the UCI population.

The fidelity audit found:

- age and the income proxy were relatively close;
- education, utility reliability, savings buffer and debt-to-inflow ratio had high KS warnings;
- correlation matrices were different;
- DCR mean was 0.4864 and the fifth percentile was 0.2497, so the privacy proxy did not show near-copying.

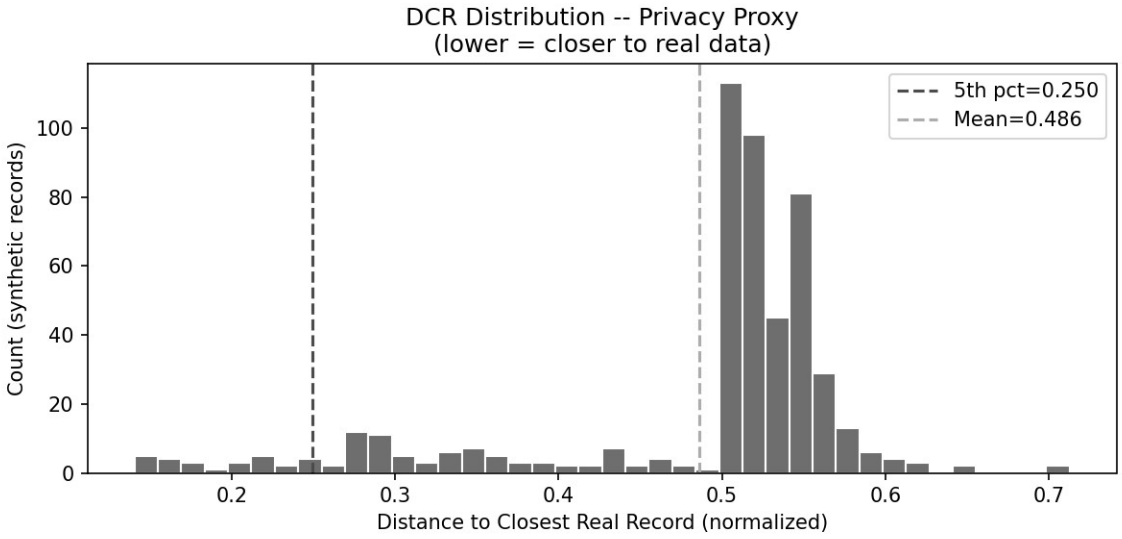


Figure 15. Distance-to-closest-record plot used as a privacy proxy. Tool: Python/Matplotlib; source: validation/step4_fidelity_audit.py.

These checks make the report more honest. They show both what worked and what still needs improvement.

9. Conclusion, Recommendations and Roadmap

IntelliCredit shows how a credit-risk project can combine individual prediction, warning rules and portfolio forecasting. On its own synthetic test set, the model performs well enough to demonstrate the idea. The dashboard makes the results easier to inspect.

The strongest lesson is that validation matters as much as model accuracy. The `data_compare` and `validation` outputs show a large gap when the synthetic model is moved to UCI data. Because of this, the project should be used for education and research, not automatic lending.

Recommended next steps

14. Collect an approved, anonymised real loan dataset with the same feature meanings.
15. Rebuild the feature crosswalk using real gig-worker transaction fields rather than imputed constants.
16. Calibrate the model separately for different borrower groups and time periods.
17. Add fairness checks using real outcomes and clear consent.
18. Compare model decisions with human review and measure approval quality, not only AUC.

Time	Action	Responsible stakeholder	Success check
Immediate	Freeze the current research prototype and document its limits	Project team	Reproducible validation run
Medium term	Collect consented, anonymised real lending outcomes	Lender and data team	Real-data calibration and subgroup report
Longer term	Pilot human-in-the-loop underwriting and monitoring	Risk and compliance team	Approval quality, default rate and fairness review

The current scope does not include real borrower consent, production security, live lending decisions or a causal impact study. These are future enhancements, not completed features.

References

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Appendix A. Extended Data Dictionary and Validation Records

Purpose	Files
Data generation	`data_generation.py`, `data/data_dictionary.txt`
Cleaning and EDA	`notebooks/01_M1_data_cleaning_preprocessing.ipynb`, `notebooks/02_M2_exploratory_data_analysis.ipynb`
Feature and model work	`notebooks/03_M3_feature_engineering.ipynb` to `05_M5_evaluation_interpretation.ipynb`
Forecasting	`notebooks/06_R2_M1_time_series_forecasting.ipynb`
Association rules	`notebooks/07_R2_M2_association_rule_mining.ipynb`
Dashboard	`dashboard/app.py`
External comparison	`data_compare/UCI_Credit_Card.csv`
Validation scripts	`validation/step1_feature_crosswalk.py` to `validation/step4_fidelity_audit.py`

The extended data dictionary is stored in `data/data_dictionary.txt`. It defines borrower, transaction, macro, engineered and target fields. The validation record is stored in `validation/results/run_summary.txt`, `experiment_results.json`, `fidelity_summary.json` and `fidelity_ks_wasserstein.csv`. These files contain the numeric results behind the tables and plots in Sections 2 and 8.

Appendix B. Additional Figures, Supplementary Analysis and Tuning

The supplementary figures include `validation_figures/ks_comparison.png`, `validation_figures/policy_comparison.png`, `validation_figures/fidelity_corr_synth.png`, `validation_figures/fidelity_corr_uci.png` and `validation_figures/calibration_tstr.png`. The model tuning notebooks record the candidate algorithms, cross-validation process, SMOTE placement and final comparison.

Model family	Main settings or check	Purpose
Logistic Regression	Tuned regularisation	Interpretable baseline
Random Forest	Tuned tree count and depth	Non-linear baseline
XGBoost	Tuned boosting and learning rate	Ranking comparison
LightGBM	Calibrated probability output	Dashboard decision support
SARIMAX	Seasonal order with macro exogenous variables	Portfolio forecast
Apriori / FP-Growth	Support, lift, Fisher filtering	Warning-rule discovery

Error analysis is reported through recall, PR-AUC, calibration, transfer AUC, KS distance and policy default rates. This avoids relying only on one headline metric.

Appendix C. Implementation Details and Dashboard User Guide

C.1 Installation and commands

```
python -m venv venv
pip install -r requirements.txt
python data_generation.py
```

```
python -m src.macro_api --offline
python -m src.portfolio_simulation
python -m src.underwriting_model
python validation/run_all.py
streamlit run dashboard/app.py
```

C.2 Dashboard user guide

26. Open the **Underwriter** tab and enter or select borrower features.
27. Read the probability of default, decision threshold and SHAP reasons.
28. Use **Portfolio** to inspect historical default rates and stress scenarios.
29. Use **Warning Rules** to see behavioural combinations and lift.
30. Use **Dataset Explorer** to filter records and check distributions.
31. Use **Combined Insights** to compare borrower PD, rule counts and macro overlays.

The saved model, rule table and portfolio files are loaded from the repository. The dashboard should not be connected to real personal information without security, consent and access controls.

C.3 Configuration and sample outputs

Shared paths, ordinal maps and random seeds are in `src/config.py`. Sample data is available in `data/sample/` and `data/sample_final_dataset.csv`. Example dashboard screenshots are in `reports/screenshots/`.

Appendix D. Tracking Board, Contributions and AI Declaration

The project tracking evidence is in `GITHUB_BOARD_SETUP.md` and `reports/project_management_and_contributions.md`. The planned board stages are Backlog, To Do, In Progress, Review/Testing and Completed. The task history includes data generation, cleaning, EDA, modelling, forecasting, association rules, dashboard work, validation and report preparation.

Team member	Contribution record	Evidence
Darun R G (CB.SC.U4CSE23215)	Data architecture, synthetic generator, feature engineering and validation	`data_generation.py`, `notebooks/03_M3_feature_engineering.ipynb`, `data/DATASET_VALIDATION_SUMMARY.md`

Team member	Contribution record	Evidence
VIMALHARIHAR S K (CB.SC.U4CSE23254)	SARIMAX forecasting, association rules and predictive modelling tuning	notebooks/04_M4_predictive_modeling.ipynb, notebooks/06_R2_M1_time_series_forecasting.ipynb, notebooks/07_R2_M2_association_rule_mining.ipynb
JUTURU NAGA ABHINAVA SAI (CB.SC.U4CSE23563)	EDA, hypothesis tests and portfolio simulation	notebooks/02_M2_exploratory_data_analysis.ipynb, src/portfolio_simulation.py
Gowtham Reddy Peddireddy (CB.SC.U4CSE23322)	Data cleaning and interactive dashboard	notebooks/01_M1_data_cleaning_preprocessing.ipynb, dashboard/app.py
D R Adithyaa Seyyone (CB.SC.U4CSE23305)	Model evaluation, SHAP, synthesis and reporting	notebooks/05_M5_evaluation_interpretation.ipynb, reports/

Generative AI supported coding suggestions, debugging, editing and formatting. Humans selected the methods, ran the experiments, checked the figures and corrected the report. The academic-originality declaration is that this report is prepared for this project, borrowed ideas and software are cited, and no private personal data is included.